

ASSIGNMENT 2: USER DEPENDENT AND INDEPENDENT ANALYSIS

CSE 572: DATA MINING

A PROJECT REPORT

Submitted by

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Introduction

In this project, an attempt has been made to develop a computing system that can understand human activities. Data is provided for eating action mixed with other unknown activities. The data consists of 8 EMG sensors for a single user. The aim of the assignment is to identify the eating activities amongst the noise. The assignment is comprised of two phases. In the first phase, we consider the new set of features obtained through PCA in Project 1. For each user, we divide 60% of the data into training set both for eating action and not eating action and the rest 40% of the data as testing set both for eating action and not eating action. Using the training set, we train three types of machines: decision tree, support vector machine and neural network. The models are then validated using the testing set and model evaluation metrics such as accuracy, precision, recall and F1 score are reported. In the second phase, we consider the new set of features obtained through PCA in Project 1. For all users, we divide 60% of the data into training set both for eating action and not eating action and the rest 40% of the data as testing set both for eating action and not eating action. Using the training set, we train three types of machines: decision tree, support vector machine and neural network. All file names mentioned in this project report here onwards are available in the project folder for your reference. The entire project was coded in Python using some of its libraries like numPy, pandas and scikit-Learn.

Project Phase I: User Dependent Analysis

The data provided is collected primarily using two sources:

- A) Myo Wristband Data: The subject was wearing the wristband and performed eating actions periodically interspersed with non-eating unknown actions. The wristband provides with i) accelerometer, ii) gyroscope, iii) orientation and iv) EMG sensors. But for the scope of this assignment we are going to only consider EMG sensors for all users. The sampling rate of EMG sensors is 100 Hz.
- B) Ground Truth Data: Video recording of the person performing eating acting actions is used for establishing ground truth data. The ground truth data is provided in the form of frame numbers where an eating action starts and ends. The video data is taken at 30 frames per second.

For each user (30 users), we have applied 6 feature extraction methods like standard deviation, mean, min, max, rms and fast fourier transform (fft). For fft there are 8 features per sample (AMP_FFT1 to AMP_FFT8). But since $AMP_FFT2 = AMP_FFT8$, $AMP_FFT3 = AMP_FFT7$ and $AMP_FFT4 = AMP_FFT6$, we have only considered AMP_FFT1 to AMP_FFT5 in our feature matrix. For each user, we have then divided 60% of the data as training for both eating and not eating actions and the rest 40% of the data as testing for both eating and not eating actions. Now, since the not eating action samples in the case of training and testing are significantly higher than eating action samples, we have selected the same number of eating and not eating action samples in training and testing sets.

In order to reduce the dimensionality of the feature matrix we performed PCA on both the training and testing sets.

PCA for Training Set:

```
[0.43221332 0.19301987 0.16599063 0.08704134 0.04775326]
92.601841270221
```

The variance for the first 5 principal components for the training set was found to be ~92%.

PCA for Testing Set:

```
[0.44761191 0.19376699 0.16523096 0.08019619 0.04561875]
93.24248127868253
```

The variance for the first 5 principal components for the training set was found to be ~92%.

PCA for Training Set is defined as 'Phase1_PCA_Training_NewFeatureMatrix.csv' and the path is:

/Akshay_Deshneni_DataMiningAssignment2

PCA for Testing Set is defined as 'Phase1_PCA_Testing_NewFeatureMatrix.csv' and the path is:

/Akshay_Deshneni_DataMiningAssignment2.

Now using the new feature matrices for training and testing sets obtained from PCA we have developed three models: Decision Tree, Neural Network and Support Vector Machine.

Decision Tree:

Using DecisionTreeClassifier method in python's scikit-Learn the decision tree has been developed.

The parameters used are : class_weight=None, criterion='entropy', max_depth= 2, max_features= 'auto' , max_leaf_nodes=None, min_impurity_decrease=0.0, min_impurity_split=None, min_samples_leaf=1, min_samples_split= 25, min_weight_fraction_leaf=0.0, presort=False, random_state= 15, splitter='best'.

The decision tree is trained using training set and then we fed the testing set for predicting the class label.

The predicted output of testing set by the model is then compared to the actual output. Model evaluation metrics such as Accuracy, Precision, Recall and F1 score have been reported below:

Decision Tree

Accuracy of Decision Tree: 0.5265763418447108

Precision of Decision Tree: 0.5194682436772138

Recall of Decision Tree: 0.7091325673920775

F1 Score of DecisionTree: 0.5996607620613076

	precision	recall	f1-score	support
Eating	0.52	0.71	0.60	120897
NotEating	0.54	0.34	0.42	120897
accuracy			0.53	241794
macro avg	0.53	0.53	0.51	241794
weighted avg	0.53	0.53	0.51	241794

Confusion Matrix:

```
[[85732 35165]
 [79306 41591]]
```

The model gave a good recall of 71% for eating actions, but further tuning of hyper parameters and optimization might be required to make the metrics better.

Neural Network (Multi-Layer Perceptron):

Using MLPClassifier method in python's scikit-Learn the neural network has been developed.

The parameters used are: activation='relu', alpha=0.001, batch_size='auto', beta_1=0.9, tol=1e-4, beta_2=0.999, early_stopping=False, epsilon=1e-08, hidden_layer_sizes=(50,) , learning_rate='adaptive', learning_rate_init=0.001, max_iter=200, momentum=0.9, n_iter_no_change=10, nesterovs_momentum=True, power_t=0.5, random_state=None, solver='adam'.

The neural network is trained using training set and then we fed the testing set for predicting the class label.

The predicted output of testing set by the model is then compared to the actual output. Also, in order to decrease the execution time of the model we have used an ensemble method BaggingClassifier available in scikit-Learn library of python. The Model evaluation metrics such as Accuracy, Precision, Recall and F1 score have been reported below:

Neural Network – Multi layer Perceptron (MLP)

Accuracy of Neural Network: 0.5168655963340695

Precision of Neural Network: 0.5139814586247565

Recall of Neural Network: 0.6200071134933042

F1 Score of Neural Network: 0.5620376931238354

	precision	recall	f1-score	support
Eating	0.51	0.62	0.56	120897
NotEating	0.52	0.41	0.46	120897
accuracy			0.52	241794
macro avg	0.52	0.52	0.51	241794
weighted avg	0.52	0.52	0.51	241794

Confusion Matrix:

[[74957 45940]

[70879 50018]]

The model gave a recall of 62% for eating actions which is lower than decision tree, but further tuning of hyper parameters and optimization might be required to make the metrics better.

Support Vector Machine:

Using SVC method in python's scikit-Learn the support vector machine has been developed.

The parameters used are: C=1.0, cache_size=200, class_weight=None, coef0=0.0,

decision_function_shape='ovr', degree=3, gamma='auto_deprecated', kernel='rbf', max_iter=-1,

probability=False, random_state=None, shrinking=True, tol=0.001, verbose=False.

The support vector machine is trained using training set and then we fed the testing set for predicting the class label. The predicted output of testing set by the model is then compared to the actual output. Also, in order to decrease the execution time of the model we have used an ensemble method BaggingClassifier available in scikit-Learn library of python. The Model evaluation metrics such as Accuracy, Precision, Recall and F1 score have been reported below:

Support Vector Machine

Accuracy of SVM: 0.5178871270585705

Precision of SVM: 0.5205762295783894

Recall of SVM: 0.4525422467058736

F1 Score of SVM: 0.48418099595564484

	precision	recall	f1-score	support
Eating	0.52	0.45	0.48	120897
NotEating	0.52	0.58	0.55	120897
accuracy			0.52	241794
macro avg	0.52	0.52	0.52	241794
weighted avg	0.52	0.52	0.52	241794

Confusion Matrix:

[[54711 66186]

[50386 70511]]

The model gave a recall of 45% for eating actions and 58% for not eating actions. So, for not eating actions it was performing better than both decision tree and neural network, but further tuning of hyper parameters and optimization might be required to make the metrics better.

So, in comparison of all the three models, decision tree has the highest recall for eating actions whereas support vector machine had the highest recall for not eating actions. Neural network gave poor recall for not eating actions and average recall for eating actions. But since the data we are trying to classify is only binary, it might be possible that the neural network (MLP) is not required. This has also been studied by many researchers from both academia and industry where for binary classification problems decision tree or support vector machines gave better results compared to complex neural networks such as a multi layer perceptron.

The entire code for Phase 1 is defined as 'Phase1.py' and the path is:

/Akshay_Deshneni_DataMiningAssignment2

Project Phase II: User Independent Analysis

The data provided is collected primarily using two sources:

- A) Myo Wristband Data: The subject was wearing the wristband and performed eating actions periodically interspersed with non-eating unknown actions. The wristband provides with i) accelerometer, ii) gyroscope, iii) orientation and iv) EMG sensors. But for the scope of this assignment we are going to only consider EMG sensors for all users. The sampling rate of EMG sensors is 100 Hz.
- B) Ground Truth Data: Video recording of the person performing eating acting actions is used for establishing ground truth data. The ground truth data is provided in the form of frame numbers where an eating action starts and ends. The video data is taken at 30 frames per second.

For each user (30 users), we have applied 6 feature extraction methods like standard deviation, mean, min, max, rms and fast fourier transform (fft). For fft there are 8 features per sample (AMP_FFT1 to AMP_FFT8). But since $AMP_FFT2 = AMP_FFT8$, $AMP_FFT3 = AMP_FFT7$ and $AMP_FFT4 = AMP_FFT6$, we have only considered AMP_FFT1 to AMP_FFT5 in our feature matrix. For all users, we have then divided 60% of the data as training for both eating and not eating actions (18 users) and the rest 40% of the data as testing for both eating and not eating actions (12 users). Now, since the not eating action samples in the case of training and testing are significantly higher than eating action samples, we have selected the same number of eating and not eating action samples in training and testing sets.

In order to reduce the dimensionality of the feature matrix we performed PCA on both the training and testing sets.

PCA for Training Set:

[0.44479209 0.19241699 0.16398943 0.08329796 0.04483856]
92.93350242061456

The variance for the first 5 principal components for the training set was found to be ~92%.

PCA for Testing Set:

[0.43172563 0.1953586 0.1783145 0.08939564 0.04602517]
94.0819524964127

The variance for the first 5 principal components for the training set was found to be ~94%.

PCA for Training Set is defined as 'Phase2_PCA_Training_NewFeatureMatrix.csv' and the path is:

/Akshay_Deshneni_DataMiningAssignment2

Now using the new feature matrices for training and testing sets obtained from PCA we have developed three models: Decision Tree, Neural Network and Support Vector Machine.

Decision Tree:

Using DecisionTreeClassifier method in python's scikit-Learn the decision tree has been developed.

The parameters used are : class_weight=None, criterion='entropy', max_depth= 2, max_features= 'auto' , max_leaf_nodes=None, min_impurity_decrease=0.0, min_impurity_split=None, min_samples_leaf=1, min_samples_split= 25, min_weight_fraction_leaf=0.0, presort=False, random_state= 15, splitter='best'.

The decision tree is trained using training set and then we fed the testing set for predicting the class label.

The predicted output of testing set by the model is then compared to the actual output. Model evaluation metrics such as Accuracy, Precision, Recall and F1 score have been reported below:

Decision Tree

Accuracy of Decision Tree: 0.5219984656693517

Precision of Decision Tree: 0.5147923652308486

Recall of Decision Tree: 0.7655734560797852

F1 Score of DecisionTree: 0.6156230721776681

	precision	recall	f1-score	support
Eating	0.51	0.77	0.62	104280
NotEating	0.54	0.28	0.37	104280
accuracy			0.52	208560
macro avg	0.53	0.52	0.49	208560
weighted avg	0.53	0.52	0.49	208560

Confusion Matrix:

```
[[79834 24446]
 [75246 29034]]
```

The model gave a good recall of 76% for eating actions. However, it was giving a poor recall of 28% for not eating actions. Further tuning of hyper parameters and optimization might be required to make the metrics better.

Neural Network (Multi-Layer Perceptron):

Using MLPClassifier method in python's scikit-Learn the neural network has been developed.

The parameters used are: activation='relu', alpha=0.001, batch_size='auto', beta_1=0.9, tol=1e-4, beta_2=0.999, early_stopping=False, epsilon=1e-08, hidden_layer_sizes=(50,) , learning_rate='adaptive', learning_rate_init=0.001, max_iter=200, momentum=0.9, n_iter_no_change=10, nesterovs_momentum=True, power_t=0.5, random_state=None, solver='adam'.

The neural network is trained using training set and then we fed the testing set for predicting the class label.

The predicted output of testing set by the model is then compared to the actual output. Also, in order to decrease the execution time of the model we have used an ensemble method BaggingClassifier available in scikit-Learn library of python. The Model evaluation metrics such as Accuracy, Precision, Recall and F1 score have been reported below:

Neural Network – Multi layer Perceptron (MLP)

Accuracy of Neural Network: 0.5280494821634062

Precision of Neural Network: 0.5246054712473501

Recall of Neural Network: 0.5980341388569237

F1 Score of Neural Network: 0.5589184247790783

	precision	recall	f1-score	support
Eating	0.52	0.60	0.56	104280
NotEating	0.53	0.46	0.49	104280
accuracy			0.53	208560
macro avg	0.53	0.53	0.53	208560
weighted avg	0.53	0.53	0.53	208560

Confusion Matrix:

```
[[62363 41917]
```

```
[56513 47767]]
```

The model gave a recall of 60% for eating actions which is lower than decision tree and recall of 46% for not eating actions. Further tuning of hyper parameters and optimization might be required to make the metrics better.

Support Vector Machine:

Using SVC method in python's scikit-Learn the support vector machine has been developed.

The parameters used are: C=1.0, cache_size=200, class_weight=None, coef0=0.0, decision_function_shape='ovr', degree=3, gamma='auto_deprecated', kernel='rbf', max_iter=-1, probability=False, random_state=None, shrinking=True, tol=0.001, verbose=False.

The support vector machine is trained using training set and then we fed the testing set for predicting the class label. The predicted output of testing set by the model is then compared to the actual output. Also, in order to decrease the execution time of the model we have used an ensemble method BaggingClassifier available in scikit-Learn library of python. The Model evaluation metrics such as Accuracy, Precision, Recall and F1 score have been reported below:

Support Vector Machine

Accuracy of SVM: 0.529415995397008

Precision of SVM: 0.5260750928672827

Recall of SVM: 0.5934790947449176

F1 Score of SVM: 0.5577480274512101

	precision	recall	f1-score	support
Eating	0.53	0.59	0.56	104280
NotEating	0.53	0.47	0.50	104280
accuracy			0.53	208560
macro avg	0.53	0.53	0.53	208560
weighted avg	0.53	0.53	0.53	208560

Confusion Matrix:

[[61888 42392]

[55753 48527]]

The model gave a recall of 59% for eating actions and 47% for not eating actions. So, for not eating actions it was performing better than both decision tree and neural network, but further tuning of hyper parameters and optimization might be required to make the metrics better.

So, in comparison of all the three models, decision tree has the highest recall for eating actions whereas support vector machine had the highest recall for not eating actions. Neural network gave average recall for not eating actions and average recall for eating actions. So, for user independent analysis using a support vector machine would be a good choice among the three.

The entire code for Phase 2 is defined as 'Phase2.py' and the path is:

/Akshay_Deshneni_DataMiningAssignment2

CONCLUSION

In this project an attempt has been made to apply feature extraction techniques to understand the data, performed principal component analysis (PCA) to reduce the dimensionality of the data and three classification models have been developed for predicting the class labels in two scenarios: user dependent analysis and user independent analysis. In comparison of both the approaches, user independent analysis was giving slightly better results than user dependent analysis. Although the scope of this project ends at developing the three models in both approaches, further study can be conducted in tuning the models to find the best possible hyper parameters. Also, since all the three algorithms were not giving very high accuracy, further research can be made to apply ensemble modelling techniques and develop a customized model that can be a combination of a decision tree, support vector machine and multi-layer perceptron.

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